



2025

U24

BOOKLET NO.

113111

**Forest Services
Physics**

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.



SECTION – A

- Q1.** Solve **any five** out of seven. **(8×5=40)**
- Q1.** (a) Explain the concept of coriolis force. Discuss its effect on a body falling freely under gravity on the earth surface. **8**
- Q1.** (b) What is chromatic aberration ? Derive the condition for achromatism for a combination of two thin lenses placed in contact. **8**
- Q1.** (c) State and derive Poisson's and Laplace's equation, mention their physical significance in solving boundary-value problem in electrostatics. **8**
- Q1.** (d) Explain the phenomenon of magnetic hysteresis in ferromagnetic materials. Show that the hysteresis energy loss per unit volume per cycle of magnetization is equal to the area of B-H loop. **8**
- Q1.** (e) Define entropy. Show that the entropy of a perfect gas increases in an irreversible process and remain constant in a reversible process. **8**
- Q1.** (f) State and explain the theorem of equipartition of energy. How is it applied to determine the specific heat of a diatomic gas ? **8**
- Q1.** (g) Explain the concepts of attenuation and material dispersion in single-mode optical fibres. **8**
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- Q2.** (a) Derive Lagrange's equation of motion for a conservative holonomic dynamical system. Use these equations to find the equation of motion and the time period of a simple pendulum. **15**
- Q2.** (b) Describe the construction and working of a Michelson interferometer. Explain how it can be used to determine the wavelength of a monochromatic source of light and the difference in wavelength between two closely spaced spectral lines. **15**
- Q2.** (c) Define an electric dipole. Derive an expression for the potential and electric field intensity at any point due to a short electric dipole. **10**
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- Q3.** (a) State Maxwell's equation for the electromagnetic field in a vacuum. Derive the wave equation for electric and magnetic field in free space and show that electromagnetic waves travel with the speed of light. **15**
- Q3.** (b) Derive Planck's law of blackbody radiation using quantum postulates. Show how it mathematically reduces to Wien's displacement law at shorter wavelengths and the Rayleigh-Jeans law at longer wavelengths. **15**
- Q3.** (c) State the Biot-Savart law. Apply it to calculate the magnetic field induction at a point located on the axis of a circular coil carrying current. **10**
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- Q4. (a) Explain the working cycle of a Diesel Engine with the help of a standard P-V indicator diagram. Derive the mathematical expression for its thermal efficiency in terms of the compression ratio and cut off ratio. 15
- Q4. (b) Set up Euler's equation of motion for a rigid body with one point fixed. Discuss the force-free motion of a symmetrical rigid body and explain the concept of precession. 15
- Q4. (c) Explain Fraunhofer diffraction due to a single slit. Derive the condition for principal maxima, secondary maxima and minima and draw the corresponding intensity distribution curve. 10

SECTION – B

- Q5. Solve **any five** out of seven. (8×5=40)
- Q5. (a) On the basis of Heisenberg's uncertainty principle, explain why electron cannot exist inside the nucleus. 8
- Q5. (b) Discuss Stern-Gerlach experiment which demonstrate the quantization of angular momentum of an atom. 8
- Q5. (c) What do you mean by nuclear reaction ? Obtain an expression for Q-Value of nuclear reaction. 8
- Q5. (d) Distinguish between metal, semiconductor and insulator on the basis of band theory of solids. 8
- Q5. (e) Explain the construction and working of a pH meter. State its any two applications. 8
- Q5. (f) The mass of $^{35}_{17}\text{Cl}$ is 34.9800 amu. Calculate the mass defect.
If $m_p = 1.00728$ amu
 $m_n = 1.00866$ amu. 8
- Q5. (g) Calculate the height of a blood column in a capillary tube of radius 0.08 mm.
If surface tension of blood = 0.055 N/m
Angle of contact = $\cos \theta \cong 1$
Density of blood = 1060 kg/m³
Acceleration due to gravity = 9.81 m/s². 8



- Q6. (a) Obtain general time independent differential wave equation in terms of wave function ψ .
Hence derive Schrodinger's time independent wave equation for a free particle. 15
- Q6. (b) What do you mean by Nuclear Magnetic Resonance (NMR). Give an experimental arrangement for the study of NMR. Give any two applications of NMR. 15
- Q6. (c) Explain first law and second law of thermodynamics in Biological systems. 10
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- Q7. (a) Derive an expression for semi-empirical mass formula (SEMF) in nuclear binding energy.
Hence, explain various terms in SEMF equation. 15
- Q7. (b) What is an Operational Amplifier (Op-Amp) ? Explain the working of Op-Amp as an integrator with its input and output waveform. 15
- Q7. (c) State and explain physico-chemical properties in biological systems. 10
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- Q8. (a) What is a free particle ? Obtain an expression for the energy of a free particle confined in a rectangular one dimensional box of length l . 15
- Q8. (b) State and explain first and second De Morgan's theorem using truth table with three inputs A, B and C. 15
- Q8. (c) What do you mean by rigid diatomic molecules ? Give examples. Show that the moment of inertia of the diatomic molecule about an axis passing through the centre of mass is given by $I = \mu.R^2$, where $\mu \rightarrow$ Reduced mass, $R \rightarrow$ Bond length. 10
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